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ORIGIN OF *NICOTIANA TABACUM* DETECTED BY PRIMARY STRUCTURE OF FRACTION I PROTEIN

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SUMMARY

Nicotiana tabacum is believed to have arisen after hybridization of *Nicotiana sylvestris* with a species in the Tomentosae section of the genus *Nicotiana.* Recent biochemical experiments have confirmed the conclusions from previous cytogenetic studies that *N. sylvestris* was the maternal parent and have indicated that *Nicotiana tomentosiformis* was the paternal parent. However, these studies did not take into account the possibility that a new species of *Nicotiana*, called K-12, discovered in South America in 1968, could also have been one of the parents. Fraction I proteins were purified from *N. tabacum* and its putative progenitors, and separated into large and small subunits. Chymotryptic peptides of each subunit were analyzed by ion exchange column chromatography with a gradient elution system. Among 38 resolved peaks of the large subunits, 2 peaks were found to be different among the putative species. Since only *N. sylvestris* showed an identical chromatogram with *N. tabacum*, *N. sylvestris* was concluded to be the maternal progenitor, as the genetic information for the large subunit of Fraction I protein was known to be inherited by the cytoplasmic mode. On the other hand, the small subunit of Fraction I protein is inherited by the Mendelian mode and therefore *N. tabacum*, an allopolyploid, could be expected to contain two types of small subunits, one derived from *N. sylvestris* and the other from a paternal progenitor. *N. sylvestris* lacks two of the 25 chymotryptic peptides of the small subunit of *N. tabacum*. Among 3 putative paternal progenitors, these two peaks appeared only in *N. tomentosiformis*, but not in *Nicotiana otophora* or K-12. Thus, *N. tomentosiformis* was concluded to be a paternal progenitor of *N. tabacum*. The conclusion was verified by comparing chymotryptic peptides of small subunits from three amphidiploids of *N. sylvestris* crossed with *N. tomentosiformis*, *N. sylvestris* crossed with *N. otophora* and *N. sylvestris* crossed with K-12. The analytical results showed that only the progeny of *N. sylvestris* crossed with *N. tomentosiformis* contained the same small subunits as *N. tabacum*.

INTRODUCTION

The origin and evolution of *Nicotiana tabacum* has been a debatable subject for many years. Being the tobacco of commerce, *N. tabacum*, an allopolyploid, is the only one out of more than 60 *Nicotiana* species which is cultivated around the world. It is generally believed that interspecific hybridization with subsequent amphiploidy as well as gene recombination played an important role in its evolution. Until very recently, it was not possible to decide between two hypotheses on the origin of *N. tabacum* ($n = 24$). Both agree that *N. sylvestris* was one of the ancestors, and Cameron [1] obtained evidence that *N. sylvestris* contributed cytoplasm to the hybrid from which *N. tabacum* was evolved. Clausen [2] and Kostoff [3] believed that *Nicotana tomentosiformis* ($n = 12$) was the other ancestor. However, Goodspeed [4] favored *Nicotiana otophora* ($n = 12$) as the other parent because of its geographical distribution together with *N. sylvestris* ($n = 12$) in South America whereas *N. tomentosiformis* occurs in an area where *N. sylvestris* has not been found. Since these hypotheses were presented, a new species of *Nicotiana* was found in South America in 1968. The plant, called K-12 ($n = 12$), closely resembles *N. tomentosiformis* or *N. otophora* cytologically and is now being presented as a new species of the Tomentosae section of the genus *Nicotiana* [5]. An important fact is that K-12 was collected in the area where *N. sylvestris* is also found. Therefore, it could be asked whether K-12 is the progenitor of *N. tabacum* instead of the previously proposed species.

Recently, several biochemical approaches to the problem of the origin of *N. tabacum* have been reported. From a gel electrophoretic analysis of isoenzyme patterns of peroxidase, catalase and six other enzymes, Sheen [6] favored the conclusion that *N. sylvestris* and *N. tomentosiformis* were the likely progenitors of *N. tabacum.* Gray et al. [7] confirmed the conclusion, from an analysis by isoelectric focusing of the polypeptide composition of Fraction I protein. They also supported Cameron's result that *N. sylvestris* contributed the cytoplasm, so that it was the maternal progenitor of *N. tabacum.* However, these analyses had not included K-12 as a putative progenitor. Furthermore, the analysis of tryptic or chymotryptic peptides sometimes gave more precise information on the characteristic features of Fraction I proteins of different species than electrofocusing. For example, Fraction I proteins of *Nicotiana benavidesii* and *Nicotiana paniculata* were indistinguishable from those of *N. otophora* and *N. tomentosiformis* respectively, by electrofocusing, whereas the former pair could be clearly distinguished from the latter pair by the analysis of chymotryptic peptides (Wildman, S. G., personal communication and Kawashima, N., and Tanabe, Y., unpublished data). Consequently, the present study was designed to re-examine the entire question of the origin of *N. tabacum* by comparing the chymotryptic peptides of the Fraction I proteins contained in the putative ancestors including the K-12 not previously considered in other publications.

MATERIALS AND METHODS

Plants

Seeds of *N. tabacum, N. sylvestris, N. tomentosiformis, N. otophora* and K-12 were derived from self-pollinated flowers at the Iwata Tobacco Experiment Station. Seeds of the F_1 hybrid plants from *N. sylvestris* crossed with *N. tomentosiformis, N. sylvestris* crossed with *N. otophora* and *N. sylvestris* crossed with K-12, were kindly supplied by Dr. Y. Ohashi at the Utsunomiya Tobacco Experiment Station. All plants were grown in a greenhouse and the leaves were harvested 2–3 months after the germination of the seeds.

Purification and further treatment of Fraction I protein

Each 200 g of leaves was homogenized with 400 ml of ice cold Tris buffer (0.05 M, pH 7.4, including 1.0 M NaCl, 0.001 M EDTA, 0.002 M $MgCl_2$ and 0.08 M β-mercaptoethanol) using a Waring blender. Saturated ammonium sulfate solution was added to the homogenate with gentle stirring up to 30% saturation. After the solution had stood for 30 min at 5 °C, the dark green precipitate was centrifuged down. Saturated ammonium sulfate was added to the supernatant to 45% saturation; after this had stood for 1 h, the precipitate was collected by centrifugation and dissolved in about 30 ml of Tris buffer A. About 2 g (5% w/v) insoluble polyvinylpyrrolidone (Kanto Kagaku Co.) was added to the solution. After gentle stirring for 3 min, polyvinylpyrrolidone was removed by centrifugation. The pale yellow supernatant was then transferred into either a collodion or a cellophan dialysis tube and the conical end of the bag submerged in 1 l of Tris buffer B (0.025 M, pH 7.4, including 0.001 M EDTA). Fine crystals of Fraction I protein usually appeared in 1–2 days. The crystals, collected by centrifugation at 5000 $\times$ *g* for 10 min, were dissolved in about 20 ml of Tris buffer C (0.025 M, pH 7.4, containing 1 M NaCl). The solution was centrifuged at 17 000 g for 10 min to remove a small amount of insoluble contaminant. The clear supernatant was again transferred into the dialysis bag and the crystallization repeated as before. After 4 recrystallizations, the protein became gel electrophoretically pure and no further increase in specific activity for ribulose 1,5-bisphosphate carboxylase was observed upon further recrystallization. The yield was 200–400 mg from 200 g fresh leaves.

Carboxymethylation of Fraction I protein was performed according to the procedure of Kung et al. [8] but with 20-fold magnification of the scale.

Dissociation and separation of subunits of the carboxymethylated Fraction I protein was performed by Sephadex G-200 column chromatography in alkaline medium as previously reported [9]. Either large or small subunits in alkaline medium were precipitated by adjusting the medium to pH 5.5 with M HCl. The precipitates were washed with a small volume of distilled water, and then dried by lyophilization.

Chymotryptic digestion of subunits and analysis of the peptides

25 mg of large or 10 mg of small subunit was suspended in 1.5 ml of 0.2 M sodium phosphate buffer, pH 7.4, and 0.1 ml of α-chymotrypsin (Worthington) was added (2.5 and 1.0 mg/ml in 0.2 M sodium phosphate, pH 7.4, for large and small subunits, respectively). After digestion for 4 h at 37 °C, another 0.1 ml of α-chymotrypsin was added and the digestion continued for another 6 h. Then 0.5 ml of M HCl was added, to make the pH 2.2. The reaction mixture was then kept at −20 °C until used. More than 95% of the large and all of the small subunit were solubilized by the digestion.

The soluble peptides were analyzed on a column of strongly acid resin, and a constant volume gradient (0.2–2.0 M sodium concentration) was used for the chromatography. The analytical procedures were the same as those used for the tryptic peptide analysis in a previous report [10]. The results showed great reproducibility in experiments repeated 3 times with different batches of either the large or the small subunit of *N. tabacum*.

RESULTS

Chymotryptic peptides of large subunits to search for the maternal progenitor of N. tabacum

It has been reported that the genetic code for the large subunit of Fraction I protein is inherited solely from the maternal parent in all interspecific hybrid plants so far examined [11, 12]. Therefore the primary structure of the large subunit must be a good phenotypic marker for identifying the maternal progenitor of *N. tabacum*.

Fig. 1(A) shows a chromatogram of the chymotryptic peptides of the large subunit of *N. tabacum*. 38 peptides are clearly resolved. Each peak has a characteristic

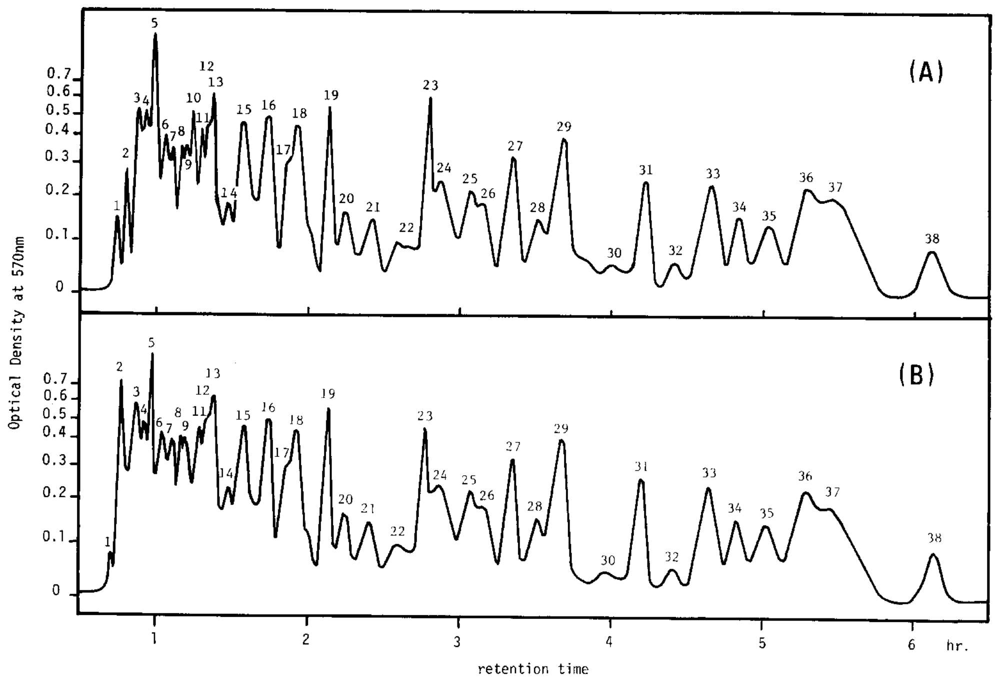

Fig. 1. Column chromatograms of chymotryptic peptides derived from the large subunit of Fraction I protein from *N. sylvestris* (A) and *N. tomentosiformis* (B). Progeny of *N. tabacum* and *N. sylvestris* crossed with *N. tomentosiformis* showed the same chromatogram as (A). *N. otophora* and K-12 showed the same chromatogram as (B).

elution volume and ratio of ninhydrin color intensity at 570 and 440 nm. The chromatograms of the large subunit of *N. sylvestris* and the hybrid of *N. sylvestris* and *N. tomentosiformis* were identical to that of *N. tabacum*. In contrast, two clear differences were found between the chromatograms of *N. sylvestris* and *N. tomentosiformis*. *N. tomentosiformis* lacks peak 10 (Fig. 1(B)). Another characteristic difference was observed in the position of peak 2. Peak 2 of *N. sylvestris* is about half the height of that of *N. tomentosiformis*. In terms of ninhydrin color, peak 2 of *N. sylvestris* had a larger A_{570nm}/A_{440nm} ratio than peak 2 of *N. tomentosiformis* (7.5 compared with 4.0). The large subunit of *N. sylvestris* could be distinguished from that of *N. tomentosi-*

formis by these differences. The large subunit of *N. otophora* or K-12 gave identical chromatograms to that of *N. tomentosiformis*. The results indicate that only *N. sylvestris* could have contributed the large subunit in *N. tabacum* and therefore was the maternal parent in the original hybridization.

Chymotryptic peptides of small subunit to search for the paternal progenitor of N. tabacum

It has been proved that genetic information for synthesis of the small subunit is transferred by pollen to egg cells during plant hybridization [13]. Recently, Sakano et al. [12] confirmed the above fact and added more information, to the effect that the amphidiploid of interspecific hybrids always contains two types of small subunit derived from both the maternal and paternal parents. *N. tabacum*, being an allopoly-

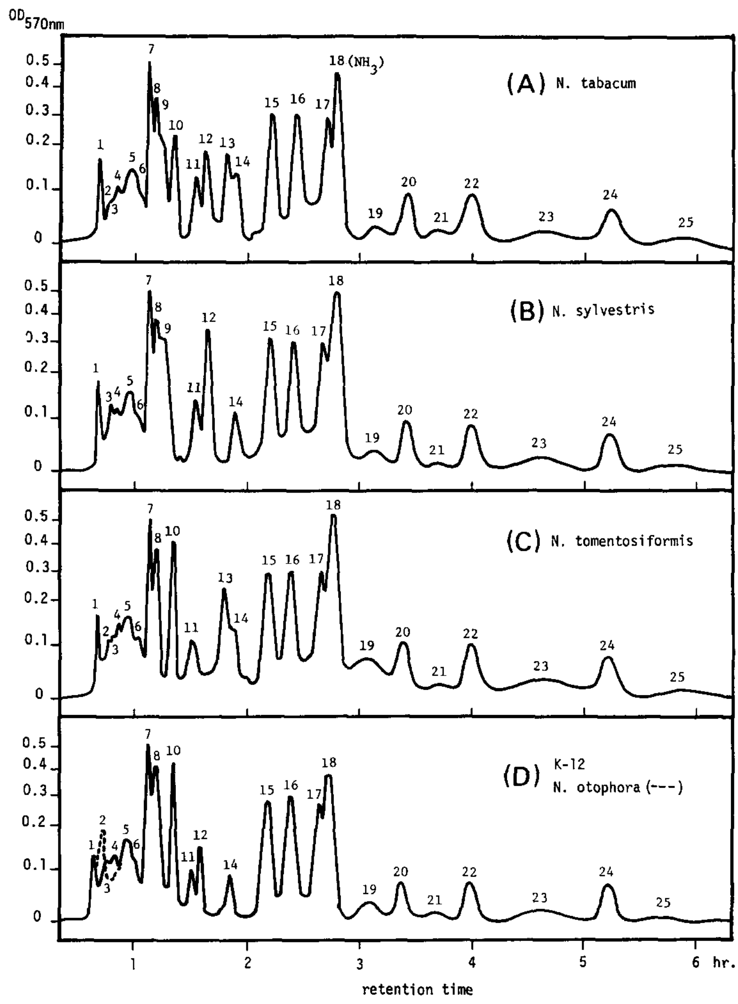

Fig. 2. Column chromatograms of chymotryptic peptides derived from the small subunits of Fraction 1 protein from *N. tabacum* and its putative progenitor species.

ploid, contains two different kinds of small subunits derived from the parents of a progenitor during the original hybridization which led to its evolution as a new species of *Nicotiana*.

As shown in Fig. 2, the chymotryptic peptides of the small subunit of *N. tabacum* are resolved into 25 peaks on a chromatogram. In comparison the small subunit of *N. sylvestris* lacks peaks 10 and 13 (Fig. 2(B)). Since the chymotryptic peptide analysis of the large subunits showed that *N. sylvestris* would have had to be the maternal parent of *N. tabacum*, then *N. sylvestris* must also have contributed one of the two small subunits of *N. tabacum*. Consequently, the paternal parent would be expected to contain the type of small subunit which possesses peaks 10 and 13 as inherent components of the primary structure. Among the three putative paternal

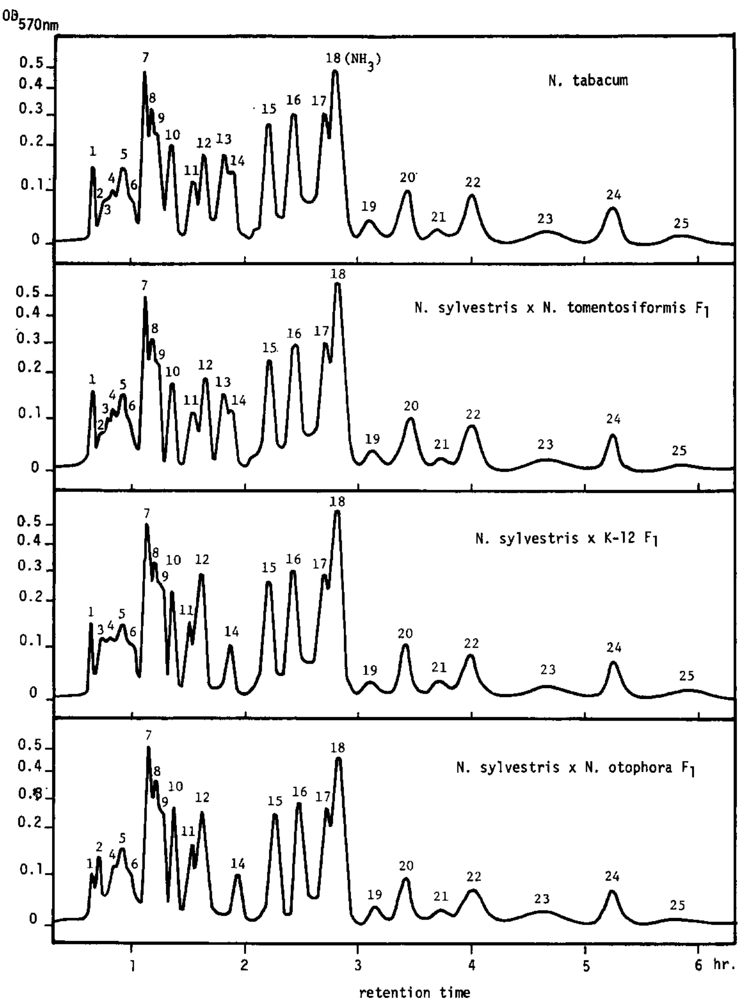

Fig. 3. Column chromatograms of chymotryptic peptides derived from the small subunits of Fraction I protein from amphidiploids of *N. sylvestris* crossed with *N. tomentosiformis*, *N. sylvestris* crossed with K-12 and *N. sylvestris* crossed with *N. otophora*.

progenitors, only *N. tomentosiformis* contains both peaks 10 and 13 (Fig. 2(C)). *N. otophora* and K-12 contain peak 10, but both lack peak 13 (Fig. 2(D)). The results clearly show that evolution of *N. tabacum* did not involve either *N. otophora* or K-12 as a paternal parent, and only *N. tomentosiformis* is eligible to be the paternal progenitor of *N. tabacum*.

To substantiate further the above conclusion, small subunits were prepared from three kinds of experimentally produced hybrid plants, of *N. sylvestris* crossed with *N. tomentosiformis*, *N. sylvestris* crossed with K-12 and *N. sylvestris* crossed with *N. otophora*, and their chymotryptic peptides were compared with those of *N. tabacum*. As shown in Fig. 3, the small subunit of the amphidiploid of *N. sylvestris* crossed with *N. tomentosiformis* shows exactly the same chromatographic profile of chymotryptic peptides as that of *N. tabacum*. On the other hand, the amphidiploid of *N. sylvestris* crossed with *N. otophora* or *N. sylvestris* crossed with K-12 still lacks peak 13 on the chromatogram. Thus it is concluded that *N. sylvestris* crossed with *N. tomentosiformis* is the only combination by which *N. tabacum* could have arisen in the original hybridization.

Ratio of the two small subunits in N. tabacum

Among 25 chymotryptic peptides of the small subunit, *N. sylvestris* lacks peaks 10 and 13, whereas *N. tomentosiformis* lacks peaks 9 and 12. By calculating the peak areas of these characteristic peptides for each subunit, the proportion of the two small subunits in *N. tabacum* was estimated. The results are shown in Table I. Regarding the peak area of *N. tabacum* as a standard of comparison, *N. sylvestris* contains twice

TABLE I

COMPARISON OF THE AREAS OF THE PEAKS OF CHARACTERISTIC PEPTIDES DERIVED FROM SMALL SUBUNITS OF FRACTION I PROTEIN

	Peptide number			
	9	10	12	13
N. tabacum (10 mg)	1	1	1	1
N. sylvestris (10 mg)	2	0	2	0
N. tomentosiformis (10 mg)	0	2	0	2
N. sylvestris × *N. tomentosiformis* (10 mg)	1	1	1	1
N. sylvestris (5 mg) + *N. tomentosiformis* (5 mg)	1	1	1	1

Numbers are given as relative value to the peak area of *N. tabacum* (cf. Figs. 2 and 3).

as much peak 9 and peak 12. On the other hand, *N. tomentosiformis* contains twice as much peak 10 and peak 13. The results suggest that *N. tabacum* contains two small subunits in a 1:1 ratio. Actually, a 1:1 mixture of the two small subunits from *N. sylvestris* and *N. tomentosiformis* showed the same quantities of these peptide areas as those of *N. tabacum*. This fact may indicate an equal contribution of maternal and paternal nuclear genes in *N. tabacum*. If this is the case, we may suspect that only one hybridization had occurred during the origination of *N. tabacum* with no more crossings with either *N. sylvestris* or *N. tomentosiformis*.

DISCUSSION

As described in the Introduction, K-12 has been proved cytologically to be a new species of *Nicotiana* belonging to the Tomentosae section [5]. The present results support the proof biochemically, since the structure of the small subunit of K-12 is different from either of the other species in the Tomentosae section. Besides the biochemical confirmation, the present results permit a reasonable speculation of the evolution of the species in the Tomentosae section. As shown in Fig. 2(C), the small subunit of *N. tomentosiformis* differs in two peptides (peaks 12 and 13) from that of K-12, whereas it differs in at least 4 peptides (peaks 2, 4, 12 and 13) from that of *N. otophora*. On the other hand, K-12 differs in 2 peptides (peaks 2 and 4) from *N. otophora*. According to Goodspeed [4], species of the Tomentosae section developed by mutation from an ancestor species of *N. otophora*. If this is the case, it may be suspected that *N. tomentosiformis* and K-12 were evolved from *N. otophora* in a similar way. Or, a more attractive possibility in our eyes is that *N. tomentosiformis* evolved from *N. otophora* via K-12. Further resolution of these speculations may result from comparative analysis of amino acid sequences of the small subunits among these species, and our experiments are proceeding along these lines.

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